MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL, FOUR WALL HEADER, RIGHT ANGLE (.100 SPACING) WITH POLARIZING FEATURE, SOLDERLESS WRAPPOST OR PWB TERMINATION

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the header connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-83503(USAF).

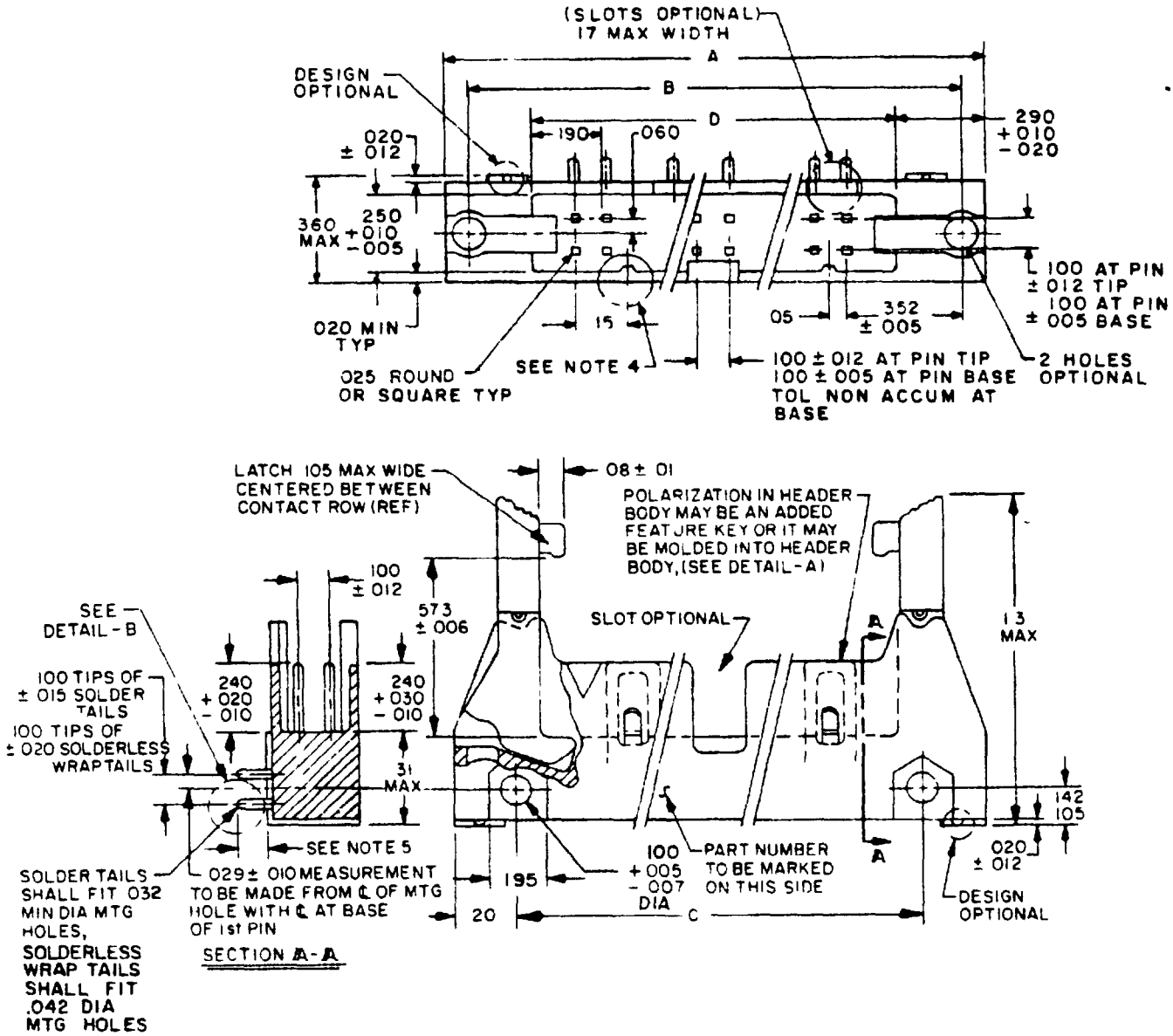
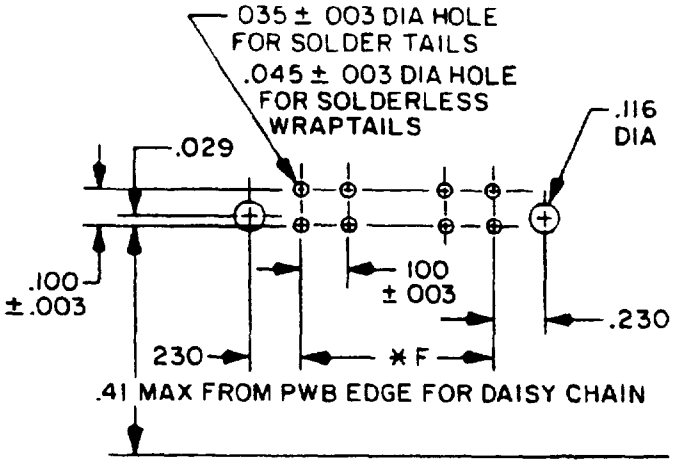


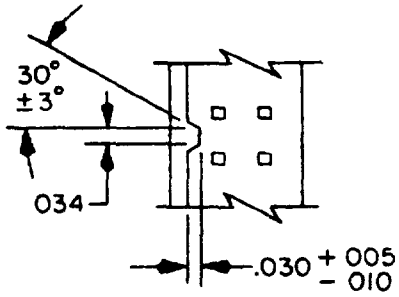
FIGURE 1. Dimensions and configurations

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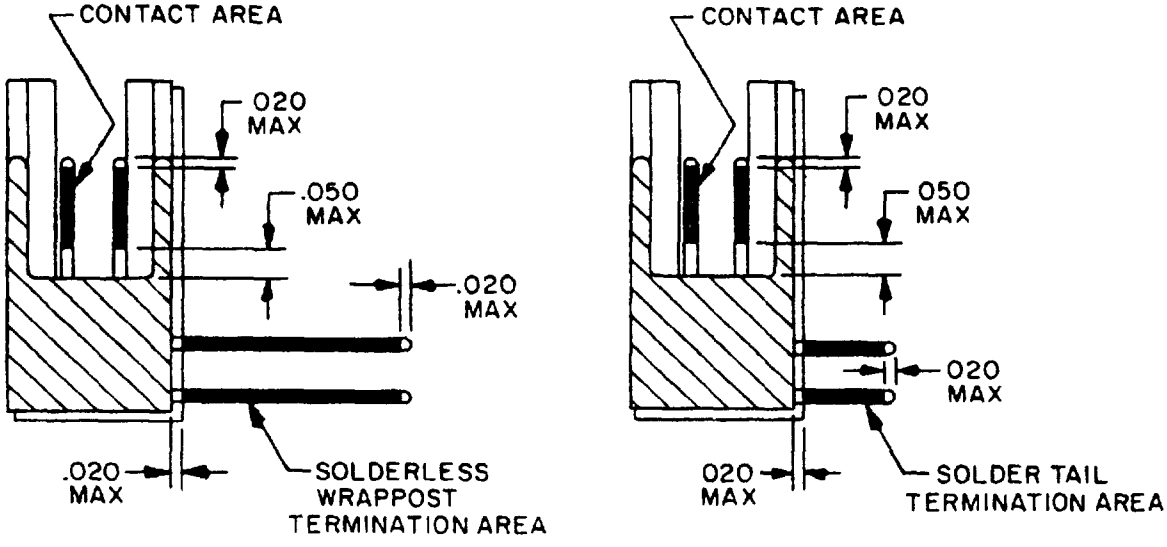
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RECOMMENDED
MOUNTING HOLE PATTERN



POLARIZATION DETAIL
DETAIL-A



FINISH LOCALIZATION AREA
DETAIL-B

FIGURE 1 Dimensions and configuration - Continued

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Inches	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.030	.76	.100	2.54	.20	5.1
.005	.13	.034	.86	.105	2.67	.230	5.84
.006	.15	.035	.89	.116	2.95	.240	6.10
.010	.25	.045	1.14	.125	3.18	.250	6.35
.01	.3	.05	1.3	.142	3.61	.290	7.37
.012	.30	.050	1.27	.15	3.8	.31	7.9
.015	.38	.060	1.52	.152	3.86	.352	8.94
.020	.51	.062	1.57	.17	4.3	.360	9.14
.025	.64	.08	2.0	.178	4.52	.41	10.4
.029	.74	.089	2.26	.190	4.83	.573	14.55
				.195	4.95	.610	15.49
						1.3	33.0

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.01 (.3 mm) for two place decimals.
4. The polarizing feature on this end is non-existent on the 10 position connector and 14 position connector.
5. For .062 PWB pin length shall be .089 minimum and .116 maximum, for .125 PWB, pin length shall be .152 minimum and .178 maximum, and for solderless wrappost, pin length shall be .610 ±.015.

FIGURE 1 Dimensions and configuration - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number	No. of pins	Dimensions 1/ A	B	C	D +.035 -.000	E	F	Termination, straight contact type
-01	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	.062 PWB
-02	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	.125 PWB
-03	10	1.26 (32.0)	1.10 (27.9)	.86 (21.84)	.690 (17.53)	4	.400 (10.16)	Wrappost
-04	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.062 PWB
-05	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	.125 PWB
-06	14	1.46 (37.1)	1.30 (33.0)	1.06 (26.92)	.890 (21.34)	6	.600 (15.24)	Wrappost
-07	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.062 PWB
-08	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	.125 PWB
-09	16	1.56 (39.6)	1.40 (35.6)	1.16 (29.46)	.990 (25.15)	7	.700 (17.78)	Wrappost
-10	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	.062 PWB
-11	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	.125 PWB
-12	20	1.76 (44.7)	1.60 (40.6)	1.36 (34.54)	1.190 (30.23)	9	.900 (22.86)	Wrappost
-13	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	.062 PWB
-14	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	.125 PWB
-15	26	2.06 (52.3)	1.90 (48.3)	1.66 (42.16)	1.490 (37.85)	12	1.200 (30.48)	Wrappost
-16	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	.062 PWB
-17	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	.125 PWB
-18	34	2.46 (62.5)	2.30 (58.4)	2.06 (52.32)	1.890 (48.01)	16	1.600 (40.64)	Wrappost
-19	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	.062 PWB
-20	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	.125 PWB
-21	40	2.76 (70.1)	2.60 (66.0)	2.36 (59.94)	2.190 (55.63)	19	1.900 (48.26)	Wrappost
-22	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	.062 PWB
-23	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	.125 PWB
-24	50	3.26 (82.8)	3.10 (78.7)	2.86 (72.64)	2.690 (68.33)	24	2.400 (60.96)	Wrappost
-25	60	3.76 (95.5)	3.60 (91.4)	3.36 (85 34)	3.190 (81.03)	29	2.900 (73.66)	.062 PWB
-26	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81.03)	29	2.900 (73.66)	.125 PWB
-27	60	3.76 (95.5)	3.60 (91.4)	3.36 (85.34)	3.190 (81 03)	29	2.900 (73.66)	Wrappost
-28	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	.062 PWB
-29	64	3.96 (100 6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	.125 PWB
-30	64	3.96 (100.6)	3.80 (96.5)	3.56 (90.42)	3.390 (86.11)	31	3.100 (78.74)	Wrappost

1/ Metric equivalents are in parentheses

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REQUIREMENTS.

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Temperature range: -55°C to + 120°C

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Phosphor bronze or brass per CDA260, spring temper; or copper alloy CA-725 per ASTM-B-122.

Contact plating:

Contact area: Gold

Solderless wrappost termination area: Per MIL-STD-1130.

Solder tail termination area: Gold or tin-lead.

Housing: Glass filled polyester per MIL-M-24519, type GPT-15F, GPT-20F or GPT-30F.

Accessories: Parts may be constructed of polyester per MIL-M-24519 or other materials specified in MIL-C-83503, provided these parts do not come in contact with current carrying components.

Polarization key: Shall be permanently attached to header (added feature key shall be bonded to header).

Mating connector: See MIL-C-83503/7.

Solderless wrappost contacts. Geometry parallelism and tip configuration shall be in accordance with MIL-STD-1130.

Solderless wrappost contact wire accommodation: 26 or 28 AWG solid.

Dielectric withstanding voltage (at sea level): 500 V rms, minimum.

(at altitude): 200 V rms, minimum.

Cable retention: Not applicable.

Contact retention: Not applicable.

Connector-cable flexing: Not applicable.

Contact resistance 50 milliohms, maximum.

Latch retention: The latch shall retain the mating connector when the cable retention force is applied to the mating connector cable. The latch shall also function as an ejector to eject mating connector.

Marking.

Part number: M83503/24-(dash number from table I).

Contact identification: Molded triangle "Δ" or notch to show number 1 contact, or raised numbers to show position number "1".

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Review activities:
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